

PRAVEEN DAS

Full-Stack Developer (MERN)

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PROFESSIONAL SUMMARY

Full-Stack Developer specializing in the **MERN stack**, **TypeScript**, and backend engineering. Builds scalable web applications and APIs with **Node.js**, **Express.js**, **React.js**, **MongoDB**, and **PostgreSQL**. Experienced in **WebSockets**, **Redis**, **Kafka**, authentication and authorization, microservices, event-driven architecture, **AWS**, **Docker**, **CI/CD**, and **Generative AI/RAG**. Focused on reliable backend services, real-time systems, data modeling, and production-oriented application architecture.

EDUCATIONS

St Thomas College | Palai

Bachelor's Degree

- **Sustainable Agriculture**, St Thomas College, Palai

PROJECTS

Virel –Social Discovery & Travel Planning Platform | 🌐 Website

Jan 2026 - Present

Personal Project

- Architected a **location-based platform** using **React**, **TypeScript**, **Node.js**, **Express**, and **PostgreSQL/Prisma**, encompassing **15+ models** and **15+ REST API modules**.
- Engineered a high-throughput feed with **Redis Sorted Sets** and pipelined MGET caching, serving timelines from memory to eliminate database read bottlenecks.
- Built geospatial clustering using **PostGIS** bounding boxes, Supercluster, and Redis tile caching, dynamically aggregating locations across map zoom levels.
- Developed regional trending with gravity time-decay scoring and **4 BullMQ worker pipelines** for asynchronous indexing and scheduled event processing.
- Containerized the backend with multi-stage Docker and automated CI/CD using **GitHub Actions**, **AWS ECR**, and **EC2**, supported by Vitest test coverage.

Technologies / Tools Used : React.js, Node.js, Express.js, PostgreSQL, Postgis, OLAMaps, Redis, GPS Navigation, Tailwind CSS

Chatvia –Real-Time Chat Application | 🌐 Website

Jun 2024 - Present

Personal Project

- Architected a distributed chat platform using **Turborepo**, **Next.js**, **Node.js**, **TypeScript**, **MongoDB**, and **Docker**, comprising **5 dedicated microservices**.
- Built an event-driven messaging pipeline with **Apache Kafka** and **Redis Pub/Sub**, batching consumers at **64KB/1.5s** to reduce MongoDB write saturation.
- Implemented client-side **E2EE** using Web Crypto API with **ECDH P-256**, **AES-GCM-256**, and **100K-iteration PBKDF2**, including automated key rotation.
- Designed offline-first synchronization with **Zustand**, **IndexedDB**, and **Redis**, using version/timestamp deltas to minimize conversation and receipt sync payloads.
- Integrated streaming AI with **Vercel AI SDK** and **Gemini 2.5 Flash**, utilizing Redis-resumable streams and **virtua** virtualization for **60 FPS rendering**.

Technologies / Tools Used : React.js, Node.js, Express.js, MongoDB, Redis, WebSockets, Kafka, Tailwind CSS

Artworld –Digital Art Gallery & Marketplace | 🌐 Website

Sep 2023 - Present

Personal Project

- Engineered a multi-vendor marketplace with **React**, **TypeScript**, **Node.js**, and **Express**, integrating Razorpay Route for KYC, payment splits, and webhook verification.
- Built an **ACID-compliant inventory engine** with PostgreSQL and Prisma, using pessimistic locking across **20+ models** to eliminate checkout race conditions.
- Implemented faceted catalog search with PostgreSQL full-text search and Prisma aggregations, paired with TanStack Query and Zustand for cache synchronization.
- Designed secure deployment with **OAuth 2.0**, ImageKit CDN optimization, Docker orchestration, and Caddy reverse proxy for production traffic management.

Technologies / Tools Used : React.js, Three.js, Node.js, Express.js, PostgreSQL

SKILLS

Languages/Frameworks :

JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js

Database/Cloud :

MongoDB, PostgreSQL, Redis, Valkey, Prisma, AWS, Firebase

Others :

REST APIs, WebSockets, Message Queues, Docker, CI/CD, Git, Tailwind CSS, Redux, Three.js